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Development and validation of a deep learning model for early detection and screening of diabetic retinopathy

Feifei Cao¹, Xitong Guo^{1*}, Meng Li¹, ShuJu Li² and Xin Peng³

Abstract

Early diagnosis and screening of diabetic retinopathy (DR) are crucial for reducing medical burdens and conserving healthcare resources. This study introduces an advanced AI-assisted recognition system designed to enhance the detection of DR lesions through innovative automatic learning methods. Central to our approach are agnostic text instruction templates, which facilitate zero-shot DR detection by integrating text embeddings with visual information. Our system performs comprehensive lesion detection by leveraging similarity mapping at both the image and patch levels, enabling it to identify a wide range of diabetic retinopathy (DR) lesions without the need for extensive annotated data. This AI-assisted system distinguishes itself from traditional fully supervised models and few-shot learning approaches by addressing the complexities of DR image annotation and safeguarding patient privacy. To validate the system's effectiveness, we conducted extensive experiments across five internal and publicly available test sets, as well as an external test set captured using smartphone devices. Our evaluation involved performance analysis of various pre-training methods, including detailed patch-level visualizations and t-SNE clustering techniques to assess the quality of feature embeddings. The results of our zero-shot experiments reveal that our system outperforms conventional transfer learning-based DR detection methods. This superiority is evident in both the pre-training and testing phases, showcasing the system's ability to deliver accurate and reliable DR lesion detection while circumventing the limitations of traditional approaches.

Clinical trial number

Not applicable.

Keywords Diabetic retinopathy, Lesion detection, Object-agnostic prompt learning, Zero-shot detection

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Introduction

Diabetic retinopathy (DR) remains one of the leading causes of preventable blindness worldwide, with profound implications for both patient quality of life and the strain it places on healthcare systems [42]. The early detection and diagnosis of DR are crucial in mitigating this burden, as timely intervention can prevent irreversible vision loss [11]. In light of this, developing advanced lesion detection systems for DR holds great potential for improving clinical outcomes by facilitating early diagnosis and treatment.

In recent years, artificial intelligence (AI) has shown substantial promise in augmenting DR detection capabilities [20]. However, traditional AI models face several challenges due to their reliance on large, labeled datasets for training. These challenges include the labor-intensive process of data annotation, the high costs associated with accurate labeling, and the difficulty in generalizing these models to new diagnostic scenarios due to overdependence on specific training data [12]. Furthermore, medical image annotation requires expert knowledge, and privacy concerns surrounding patient data present additional obstacles in developing effective AI-based solutions.

To address these limitations, zero-shot learning (ZSL) has emerged as a promising approach in medical image analysis. ZSL enables AI models to detect and classify lesions without requiring disease-specific labeled datasets by leveraging prior knowledge and identifying generalizable patterns. This method enhances the adaptability of AI models across different diagnostic scenarios, improving their ability to recognize DR lesions in varied clinical settings without extensive labeled data [43].

Recent advancements in the field have further demonstrated the effectiveness of AI in DR detection. For instance, researchers have successfully applied transfer learning with optical coherence tomography (OCT) imaging to enhance DR diagnosis, overcoming challenges posed by small and medium-sized datasets using pre-trained models [17]. Similarly [5], has employed deep learning algorithms on color fundus photography (CFP) images for DR detection, showcasing the adaptability of these models across various ethnic groups. Additionally, Google DeepMind achieved a 95% accuracy rate in DR detection using a two-stage deep neural network for both lesion segmentation and grade classification on OCT images. This underscores the robustness of their approach across different imaging modalities [16]. Despite these advances, however, these methodologies share a common reliance on large-scale, manually annotated datasets, which are costly and time-consuming to produce.

The reliance on human-annotated data introduces additional complications. Labeling quality may vary due

to the subjective nature of medical image interpretation, and handling medical data raises significant privacy concerns, necessitating stringent safeguards to protect patient confidentiality [36]. In response to these challenges, our study proposes a zero-shot learning-based approach for DR detection, which minimizes reliance on manually labeled data while addressing concerns about privacy. By leveraging the ZSL framework, our method provides a promising solution for enhancing DR lesion detection without requiring extensive, annotated datasets.

Most existing deep learning (DL) approaches focus on detecting moderate non-proliferative diabetic retinopathy (NPDR) or more severe stages of DR, which are more likely to cause vision-threatening conditions [12]. However, early-stage DR, particularly mild NPDR, is often underrepresented in screening protocols. Given the progressive and irreversible nature of DR, early detection is critical for preventing disease advancement, which significantly increases the risk of blindness if left untreated [29]. Research indicates that timely intervention, alongside effective management of key metabolic factors such as blood glucose levels, blood pressure, and lipid profiles, can prevent or even reverse the progression of mild NPDR [4].

Despite the recognized benefits of early detection, current supervised and transfer learning models remain constrained by the high costs and privacy concerns associated with medical data labeling. To overcome these challenges, we introduce an innovative AI framework inspired by advances in contrastive learning and large visual language models, such as those trained on datasets like ImageNet. Our model harnesses paired text-image medical data to pre-train both text and image encoders, using custom-designed text instruction templates specifically tailored for DR detection tasks. This framework enables zero-shot transfer learning, allowing the system to generalize across diverse diagnostic scenarios with minimal reliance on disease-specific annotations.

In this study, we present a novel AI system employing **object-agnostic prompt learning** for DR detection. This system uses **universal prompts** to bypass the need for extensive, manually annotated datasets, allowing for more efficient and scalable detection capabilities. The zero-shot learning framework enhances the model's ability to generalize across different DR detection tasks, reducing the need for specialized annotations for each dataset. By incorporating prompt learning and minimizing dependence on large-scale labeled data, this system addresses the critical challenges of medical data annotation and privacy concerns while advancing the field of AI-driven DR detection. Furthermore, the system's ability to accurately detect early-stage DR lesions, particularly Microaneurysms, marks a significant step forward in

preventing DR progression and improving patient outcomes through early intervention.

Research background

DR detection based on deep learning Deep learning (DL) models have gained substantial attention in medical image analysis due to their ability to extract intricate, hierarchical features from complex medical images. In the context of diabetic retinopathy (DR) detection, DL models play a pivotal role in automating lesion identification and diagnosis, thus reducing the diagnostic burden on healthcare professionals. The current state-of-the-art approaches for DR detection predominantly focus on **lesion area segmentation** and **transfer learning** techniques, both of which have shown promising results in detecting retinal abnormalities. **Lesion area segmentation** models can be broadly categorized into two types: **encoder-decoder architectures** and **Atrous convolution-based models**. Encoder-decoder structures, such as those used in [10, 37], encode the input image into a latent feature space and then decode it back into a segmentation map, highlighting critical lesion areas. This layered approach is effective in locating regions affected by DR. In contrast, **Atrous convolution models**, as demonstrated in [6], use dilated convolutions to capture multi-scale contextual information while maintaining computational efficiency. This technique is particularly useful for detecting small but clinically significant lesions, such as Microaneurysms, which are early indicators of DR progression.

To improve model robustness, researchers have also experimented with data augmentation techniques to diversify the training dataset. Common methods include image rotation, flipping, and adding noise [43]. While these techniques introduce variability, they often fail to generate realistic samples that reflect the full complexity of DR lesions, thereby limiting their effectiveness in improving model performance. Transfer learning is another widely adopted strategy in DR detection, particularly in addressing the scarcity of labeled medical data. For example [12], applied transfer learning by pre-training a neural network on large-scale datasets like ImageNet and then fine-tuning it on DR-specific data. This approach leverages feature representations learned in the source domain to improve DR detection in the target domain.

Two additional techniques within the domain of transfer learning are gaining traction: semi-supervised transfer learning and cross-domain adaptation. Semi-supervised transfer learning addresses the challenge of limited labeled data by leveraging large amounts of unlabeled data to enhance model training [39]. Cross-domain adaptation transfers knowledge from a well-annotated domain (e.g., OCT imaging) to a less annotated domain

(e.g., fundus photographs), which allows the model to generalize across imaging modalities [31]. However, both methods face challenges, particularly when there is a significant domain gap between the source and target datasets, leading to suboptimal feature transfer. This gap, combined with heterogeneous data distributions, poses a significant challenge for transfer learning applications in medical imaging [20].

Text-driven contrastive visual learning Contrastive learning, which involves training models to minimize the distance between similar samples while maximizing the distance between dissimilar ones, has shown great potential in various computer vision tasks [40]. In unsupervised representation learning, this method is particularly effective, enabling models to learn rich, discriminative feature representations without the need for extensive labeled datasets. In medical imaging, contrastive learning has been applied to visual language models (VLMs), which incorporate text instruction learning to enhance both the interpretability and precision of DR detection. For example, models like CONVIRT and CONVIRT-AC [43] utilize text instructions that capture target semantics but struggle to fully capture the complexity of abnormal lesion semantics. This limitation results in suboptimal detection performance. More recent models, such as WinCLIP [19] and VAND [9], introduce manually defined text instructions, enhancing the identification of specific anomalies but lacking flexibility across different lesion types.

CoOp [44] addresses some of these shortcomings by employing learnable text instructions, which enable the model to generalize better by capturing global exception semantics. However, it still struggles with capturing local exception semantics, which is essential for fine-grained anomaly detection. ANOMALYCLIP [45] further advances the field by learning text instructions specifically designed for identifying unknown objects or abnormalities, achieving superior performance in zero-shot anomaly detection and segmentation tasks. This method excels at identifying abnormal regions in medical images, even when the model has not previously encountered those anomalies during training [38].

Zero-shot anomaly detection (ZSAD) ZSAD is a state-of-the-art technique developed to identify anomalies that fall outside of the categories included in the training dataset [2]. Unlike traditional supervised learning, which relies on extensively annotated datasets, ZSAD detects anomalies based on patterns learned from normal samples or by leveraging prior knowledge. This approach is especially valuable in medical applications, where abnormal cases may be rare or difficult to annotate comprehensively. Recent advancements in **transfer learning** have significantly enhanced the performance

of ZSAD systems. For instance, models pre-trained on large-scale datasets like **BERT** and **Vision Transformer** have demonstrated strong capabilities in creating generalizable feature representations that can be applied to new, unseen data for anomaly detection [27]. These models rely on learning from vast amounts of data, enabling them to detect deviations from normative patterns that signal the presence of anomalies. In this study, zero-shot learning (ZSL) aims to leverage pre-training on semantically annotated medical imaging datasets (e.g., chest radiographs) to establish semantic relationships between images and textual descriptors. This acquired knowledge is then transferred to diabetic retinopathy (DR) fundus image analysis, enabling the model to predict novel categories despite fundus images being entirely absent from the training data - thus constituting a zero-shot learning paradigm.

CLIP, originally developed for zero-shot image recognition, has proven highly effective in pre-training for ZSAD by adapting to diverse target datasets through its integration of auxiliary data [32]. Building on CLIP's success, text instruction learning modules have been refined to improve ZSAD performance, particularly in medical image analysis. However, the manual annotation of text prompts for zero-shot image retrieval tasks can be inefficient and often results in low detection accuracy [14]. To address this, learnable linear projection techniques have been introduced, enhancing the local semantics of image visualizations and improving efficiency in anomaly detection tasks [15].

This study introduces a zero-shot learning (ZSL) framework specifically designed for diabetic retinopathy (DR) detection, aiming to improve diagnostic accuracy while minimizing dependence on large, annotated datasets. By applying ZSAD techniques and text-driven contrastive learning, the proposed system seeks to enhance the detection of DR lesions—particularly early-stage lesions such as Microaneurysms—thereby enabling earlier interventions and improving patient outcomes.

Methodology

Comparative pre-training of text and image encoders

The proposed methodology begins with **comparative pre-training** using a dual encoder framework consisting of a **text encoder** and an **image encoder** (Radford et al., [32], both based on state-of-the-art neural network architectures. The **image encoder** utilizes the **Vision Transformer (ViT)**, known for its superior performance in image classification tasks, while the **text encoder** employs **BERT** (Devlin [13],), optimized for natural language processing tasks. In this application, the text encoder is pre-trained using **ClinicalBERT**, (Alsentzer et al., [1] a domain-specific variant of BERT tailored for processing clinical text. The image encoder leverages **ResNet50**, a deep residual network architecture widely used in medical image analysis [18].

Given the highly specialized nature of medical text and the intra-class similarity of medical images, we adopt the **CLIP framework** [32], which maximizes **cosine similarity** between text and image embeddings. This ensures a closer alignment between textual descriptions of **diabetic retinopathy (DR)** lesions and their corresponding visual features. Both paired and unpaired data consisting of medical images and their associated clinical texts are utilized during training to enhance the model's ability to detect DR. For training efficiency, we adopt a **selective fine-tuning approach**: the text encoder's initial layers are frozen, and only the final six layers are fine-tuned, tailoring the encoder to DR detection while avoiding overfitting to early-stage features. This training strategy is illustrated in Fig. 1.

Construction of text prompts for DR detection

To improve the classification of different DR lesions, we developed a **text-based instruction template** specifically designed to optimize **zero-shot detection**. This approach is based on the principle that **text instructions** can help a model recognize unseen classes [34]. The **text instruction template** integrates categorical labels such

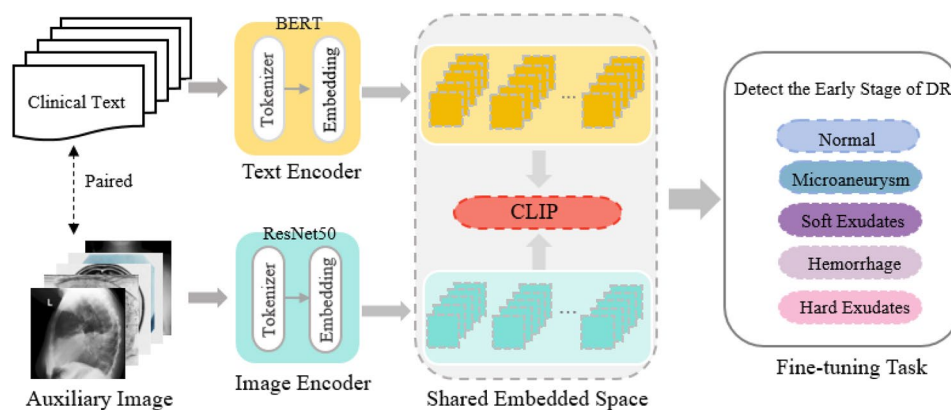


Fig. 1 Contrastive pre-training Learning

as “Microaneurysm,” which are embedded by the text encoder to enhance classification accuracy [24].

Existing templates, such as the general “A photo of a [cls]” template, have been effective in aligning text and image embeddings [25]. However, these templates often fail to capture critical **abnormal visual features**, which are essential for medical anomaly detection. To address this, we created **agnostic text instruction templates** that focus on detecting salient features of anomalies without being limited to specific categories [45]. These templates enable the model to detect abnormal features across multiple DR lesion types while maintaining generalizability to other medical conditions, thereby enhancing cross-domain **zero-shot anomaly detection (ZSAD)**.

We further improved the DR detection performance by incorporating **specific abnormal text information** into the instruction template. For example, a lesion such as a Microaneurysm or hemorrhage might be described as “A photo of a DR with a Microaneurysm.” However, such templates are limited in their ability to capture general abnormal features. To overcome this, we introduce a **learnable text instruction template**:

$$P_n = [N_1][N_2] \dots [N_m][\text{Object}],$$

$$P_a = [A_1][A_2] \dots [A_m][\text{Lesions Object}],$$

That is fine-tuned using auxiliary medical data. This process allows the model to automatically adapt the instructions to capture detailed semantic information, significantly improving its ability to differentiate between normal and abnormal images.

Learning general visual embeddings

Refinement of local visualization regions Detecting fine-grained anomalies in medical images is challenging due to the subtle visual differences between normal and abnormal tissues. To address this, we implemented the **Diagonally Prominent Attention Map (DPAM)** technique, which enhances the traditional attention mechanisms used in image encoders [45]. In standard attention mechanisms, the **Query-Key (Q-K)** attention process is central. However, we replaced it with a more refined framework that includes **Query-Query (Q-Q)**, **Key-Key (K-K)**, and **Value-Value (V-V)** self-attention mechanisms. The **V-V attention mechanism** was found to be particularly effective in capturing local visual features critical for distinguishing subtle abnormalities [15]. By focusing on the relationships between **value representations**, this method enhances the model’s ability to localize and distinguish subtle anomalies, leading to more accurate anomaly detection in medical images.

Global and local instruction optimization To maximize DR detection performance, we propose a **joint opti-**

mization strategy that enhances both **global** and **local text-visual alignment**. The global context optimization aligns text embeddings related to unknown target anomalies with visual embeddings across all relevant regions of an image. This ensures that the model captures the overall image context, improving its ability to detect both normal and abnormal areas. **Local context optimization** focuses on refining text instructions to better identify localized abnormal regions, even when these regions are rare or small. We employ **focal loss**, which emphasizes difficult-to-classify cases while reducing the impact of well-classified examples, effectively mitigating class imbalance—a common issue in medical anomaly detection where abnormal cases are rare compared to normal ones.

To further enhance the accuracy of segmentation, we incorporate **Dice loss**, a measure of the overlap between predicted abnormal regions and ground truth annotations. **Dice loss** is well-suited for medical image analysis, as it evaluates the model’s ability to capture fine details like lesion boundaries, improving anomaly segmentation performance. Figure 2 illustrates the steps in our **global/local optimization approach**, showing how these complementary strategies are combined to enhance DR detection, particularly in zero-shot tasks.

Experiment

Experiment setup

Datasets During the **pre-training phase**, we utilized two key datasets: the **MIMIC-CXR dataset** [21] and the **CBIS-DDSM dataset**. Both datasets include paired text descriptions, which aid in learning the associations between medical images and their corresponding textual information. For the **fine-tuning phase**, we employed the **MURA-1.1**, which is specifically tailored for **diabetic retinopathy (DR)** detection. The dataset was divided into training and validation sets in a **7:3 ratio** to ensure robust model evaluation and to optimize the performance of the DR detection system.

To assess the model’s **zero-shot performance**, we conducted comprehensive evaluations using both internal and external test datasets. **Internal Tests:** Several public datasets from the DR domain were used, including **DRIVE**, **ODIR-5k**, **FIRE**, **STARE**, and **CHASE-DB1**. These datasets are widely recognized in DR research and provide a broad range of DR images, which is essential for evaluating model accuracy and robustness. **External Test:** For external validation, we utilized the **IDRiD**. This dataset was obtained via smartphone imaging technology and consists of **color fundus images (CFP)**. It does not include manual annotations, making it ideal for testing the model’s performance in a real-world, less controlled setting. This enhances the generalizability of our findings. This comprehensive evaluation across diverse

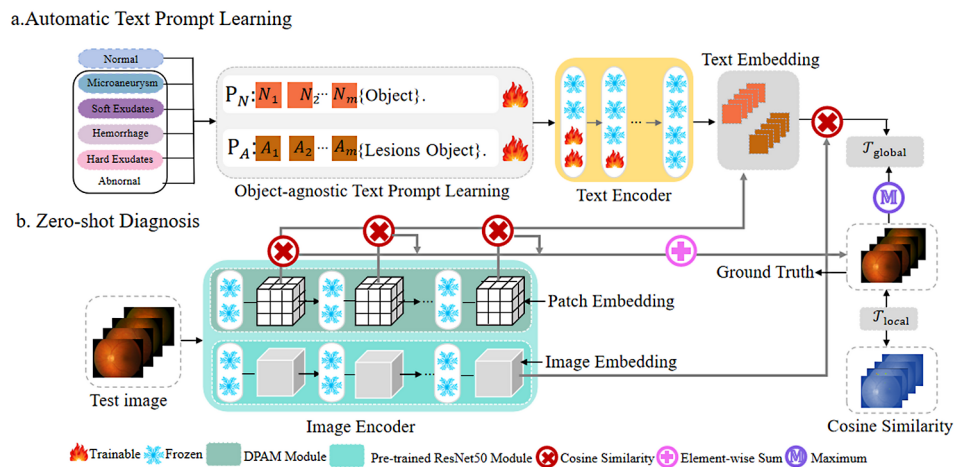


Fig. 2 Diagnostic system architecture visualization

datasets ensures that the model's performance is thoroughly assessed under a variety of conditions and data types.

Data preprocessing and evaluation metrics To ensure the quality and consistency of the input data, several preprocessing steps and evaluation metrics were employed: **Size Normalization:** All input images are resized to 336×336 pixels to maintain consistency across the dataset. This standardization is crucial for uniform data processing and model training, ensuring that all images conform to the same resolution specifications. **Image Enhancement:** Various image enhancement techniques are applied to improve the visibility of pathological features and mitigate variations in lighting and color arising from different imaging devices. Specifically, **Contrast-limited adaptive histogram equalization (CLAHE)** [30] is used to enhance contrast and reveal fine details in the images. **Color normalization** [26] is applied to standardize color variations across images, reducing discrepancies that could negatively impact the performance of deep learning algorithms. **Sensitivity** and **Specificity** are key metrics for evaluating the model's performance. **Sensitivity** measures the model's ability to correctly identify positive cases (true positives), while **specificity** measures its accuracy in identifying negative cases (true negatives). **Area Under the Receiver Operating Characteristic Curve (AUROC):** AUROC provides a comprehensive evaluation of the model's ability to distinguish between positive and negative cases across various thresholds. **Average Precision (AP)** assesses the model's performance at the image level, with a focus on the **precision-recall trade-off**. **AUPRO:** used for evaluating **image segmentation** tasks, measuring the area under the precision-recall curve for segmented regions.

Implementation details For the text encoder, we employed **BERT** [41] for pre-training, with **ClinicalBERT** [1] used to initialize the weights. **ClinicalBERT**, which is optimized for clinical language processing, is well-suited for handling medical text data. The image encoder is pre-trained using **ResNet50** [18], an architecture known for its ability to effectively extract significant features from medical images, particularly in identifying abnormal characteristics [8].

During the detection phase, a **similarity score** is used to evaluate the anomaly score. This score approaches **1** when the text embedding for anomalies is well-aligned with the **global visual embedding**, indicating a high correspondence between the identified anomaly and the visual features of the image. To achieve **fine-grained detection**, both normal and abnormal segments are integrated from all **intermediate layers** of the network. This approach facilitates a **comprehensive analysis of image features** across various levels of abstraction, leading to more accurate anomaly detection. To further enhance the model's ability to capture detailed image information, we employed the **Diagonally Prominent Attention Map (DPAM) module** [45]. The DPAM module refines visual information by incorporating **diagonal attention mechanisms**, which improve the detection of subtle and fine-grained anomalies. The **text instruction length** is set to **14 tokens**, optimizing the balance between capturing sufficient semantic detail and maintaining computational efficiency. The **depth of the learned text embedding** is set to **9 layers**, providing a robust representation of text instructions. Additionally, the **visual encoder** comprises **4 layers**, which are configured to effectively process and integrate image features, ensuring accurate anomaly detection [45].

Table 1 Zero-shot detection performance of DR_OAPL

	Dataset	Sensitivity	specificity	F1-Score	AUROC
Internal Testset	DRIVE	0.916	0.861	0.913	0.821
	ODIR-5k	0.892	0.855	0.893	0.837
	FIRE	0.848	0.904	0.881	0.917
	STARE	0.931	0.915	0.931	0.979
	CHASE-DB1	0.908	0.906	0.872	0.865
	Mean	0.899	0.888	0.898	0.884
External Testset	IDRiD	0.875	0.851	0.855	0.874
Smartphone Subset	EyePACS	0.851	0.846	0.835	0.862

Main results

Detection performance evaluation

Zero-shot detection performance To evaluate the zero-shot detection capability of the proposed DR_OAPL method, we conducted experiments using five publicly available internal diabetic retinopathy (DR) test sets and one external test set. The results of these experiments are summarized in Table 1. The DR_OAPL method exhibited competitive performance across the internal test sets. Specifically, sensitivity and specificity values exceeded 85% for all datasets except for FIRE. Sensitivity measures the proportion of actual positives correctly identified by the model, while specificity assesses the proportion of actual negatives correctly identified. These metrics indicate that DR_OAPL performs robustly in detecting DR lesions. In addition to sensitivity and specificity, we evaluated the model using the F1-Score and the area under the receiver operating characteristic curve (AUROC). Both metrics offer a comprehensive assessment of the model's performance. The F1-Score, which balances precision and recall, and AUROC, which reflects the model's ability to distinguish between classes, both remained above 80% across the datasets. These results further affirm that DR_OAPL effectively identifies fundus lesions with consistent performance. To assess the generalization performance of DR_OAPL, we compared the average detection results from the five internal test sets with those from the external test set and smartphone subset. While the detection performance on the IDRiD and EyePACS were slightly lower than on the internal sets, DR_OAPL still achieved a detection capability close to 85% across all evaluation metrics. This demonstrates the method's robustness and its ability to generalize well to new, unseen data.

Comparison with transfer model detection performance Diabetic retinopathy (DR) detection methods are varied, but mainstream approaches typically follow a standard process. Initially, these methods utilize convolutional neural networks (CNNs) pre-trained on Im-

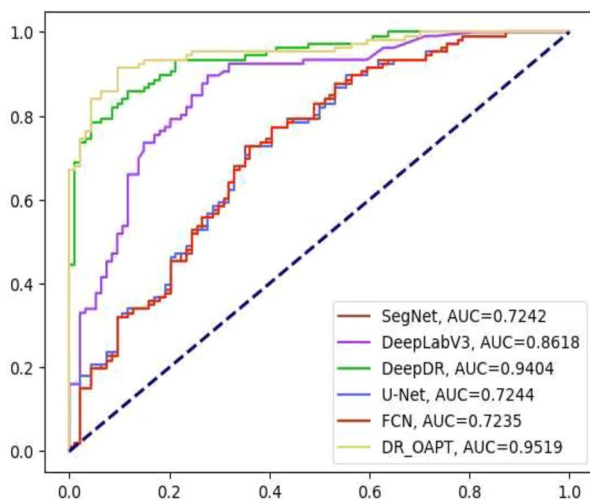
Table 2 Comparison with the transfer learning model. (Black bold indicates the best result)

	Fine-tuning		Test	
	F1-Score	AUROC	F1-Score	AUROC
FCN	0.914	0.805	0.832	0.711
SegNet	0.909	0.816	0.868	0.716
U-Net	0.904	0.814	0.870	0.787
DeepLabV3+	0.906	0.818	0.889	0.725
DeepDR	0.916	0.825	0.905	0.805
DR_OAPL	0.921	0.857	0.941	0.871

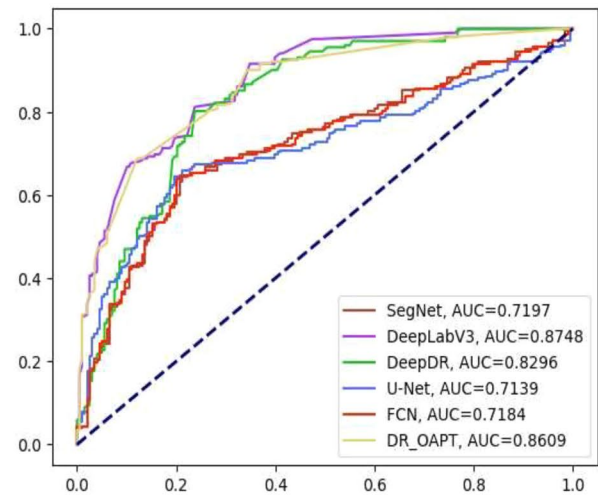
geNet to extract generic features. These features are then adapted for DR lesion detection. Notable transfer learning models that have demonstrated significant performance include Fully Convolutional Networks (FCN) [28], SegNet [3], U-Net [33], DeepLabV3 [7], and DeepDR [12].

We conducted a comparative analysis between these transfer learning models and the DR_OAPL method. The results of this comparison are detailed in Table 2, encompassing both the fine-tuning and testing phases of detection. Our findings indicate that DR_OAPL delivers superior performance across both phases. Although transfer learning models generally excel during the fine-tuning phase, they experience notable performance degradation during testing. This degradation is likely due to the models being primarily trained on the ImageNet dataset, which may not adequately capture the specific characteristics of DR data. Medical imaging, particularly for DR, often involves unique features that are not well-represented by general pre-training datasets. In contrast, DR_OAPL maintains robust performance throughout the testing phase. This enhanced performance is attributed to the incorporation of an unknown object text learning module and contrastive learning techniques. These components enable DR_OAPL to effectively capture significant abnormal features across various lesions, leading to improved accuracy in regional identification and overall detection performance.

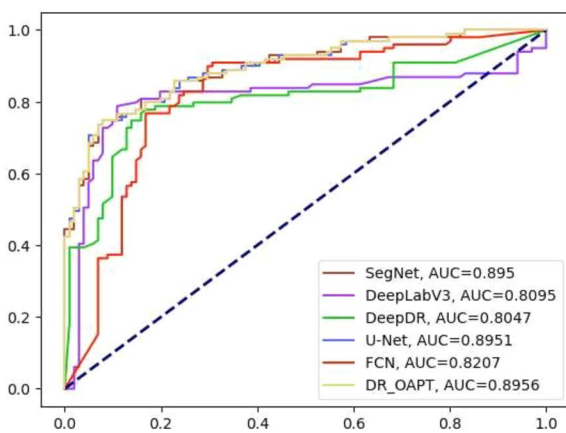
Performance of models on different lesions The detection of different lesions in diabetic retinopathy (DR) is particularly challenging due to the high similarity in features among various types. To assess the diagnostic performance of each model in detail, we visualized the Receiver Operating Characteristic (ROC) curves for different lesion types, as illustrated in Fig. 3. The results reveal that DR_OAPL excels in detecting Microaneurysms, hemorrhages, and hard exudates—critical indicators in the early stages of DR. Notably, Microaneurysms, which are often difficult to detect, achieved an Area Under the ROC Curve (AUROC) of 95.19% with DR_OAPL, while hard exudates reached an AUROC of 96.25%. This high performance can be attributed to the use of a patch-level cutting technique, which allows the model to concentrate on local features



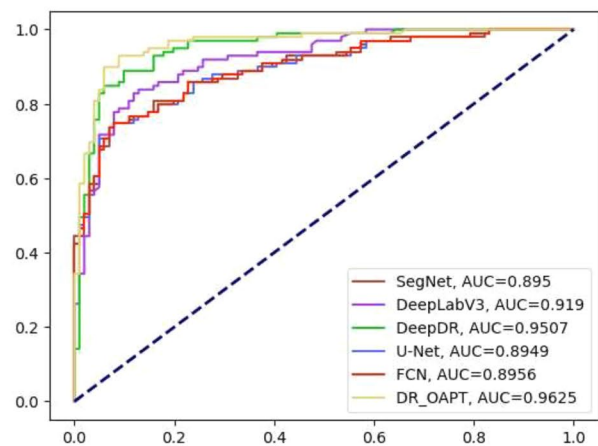
(a) Microaneurysm



(b) Cotton wool spot



(c) Hemorrhage



(d) Hard Exudates

Fig. 3 ROC curves of each model

of abnormalities, thus improving detection accuracy. In comparison, the DeepDR model also showed strong results in detecting Microaneurysms and hard exudates, with AUROC values of 94.04% and 95.07%, respectively. However, DeepDR's reliance on the ImageNet dataset for training and its use of three subnetworks for feature extraction contribute to a relatively high training cost. This makes DR_OAPT not only more effective in specific lesion detection but also more efficient in terms of training resources.

Advanced results

Segmentation visualization To assess the detection performance of the models, we visualized the local localization results for each of the six models, as depicted in

Fig. 4. The first row of the figure presents the original images of the four types of diabetic retinopathy (DR), annotated by ophthalmologists from the Second Affiliated Hospital of Ningbo Medical University. The subsequent rows, from the second to the seventh, display the segmentation outputs for each model. DR_OAPT consistently demonstrates superior performance in accurately identifying and localizing lesions. The model effectively minimizes the localization range, which aids in capturing the specific characteristics of each type of DR for precise recognition. This advantage is attributed to the integration of text instructions for unknown objects. These instructions provide detailed descriptions of the distinctive features of each anomaly,

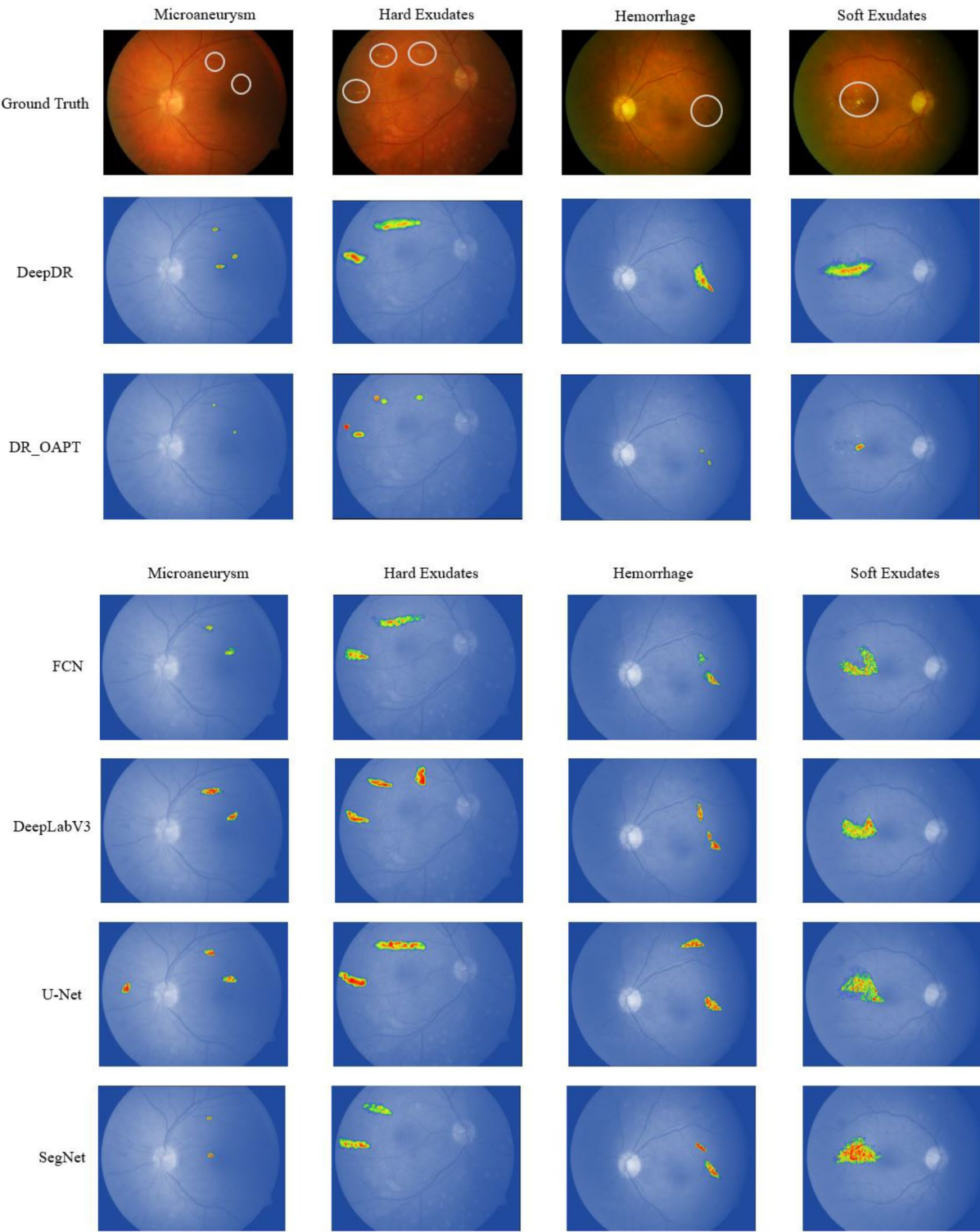


Fig. 4 Segmentation visualization of different methods

allowing the model to learn these unique patterns during the comparative pre-training phase. By employing cosine similarity loss between local and global features and the text instructions, DR_OAPL closely approximates the ground truth during both the fine-tuning and testing stages. This approach significantly enhances the accuracy of lesion localization.

Comparison of clustering effects of different pre-training methods Current mainstream diabetic retinopathy (DR) detection methods generally utilize image encoders that are pre-trained on the ImageNet dataset, which are subsequently adapted for DR-specific tasks. However, emerging research [43, 45] indicates that pre-training on medical image datasets may provide more accurate detection outcomes. To evaluate this hypothesis, we conducted a t-SNE clustering analysis comparing

pre-training on ImageNet with pre-training on medical image datasets. The results of this analysis are illustrated in Fig. 5. In our study, both ConVIRT [43] and DR_OAPL were pre-trained using the MIMIC-CXR chest radiograph dataset [21] and the CBIS-DDSM. A notable difference between these methods is that DR_OAPL incorporates an Object-Agnostic Prompt Tuning (OAPT) module. The t-SNE clustering analysis demonstrates that in-domain pre-training methods, such as those used by ConVIRT and DR_OAPL, achieve more effective clustering of diseases compared to ImageNet-based methods. Specifically, in-domain methods exhibit better clustering performance for diseases like Microaneurysms, which are often challenging to identify using general pre-training approaches. These results corroborate the findings of previous studies, underscoring the advantages of domain-specific pre-training for DR detection.

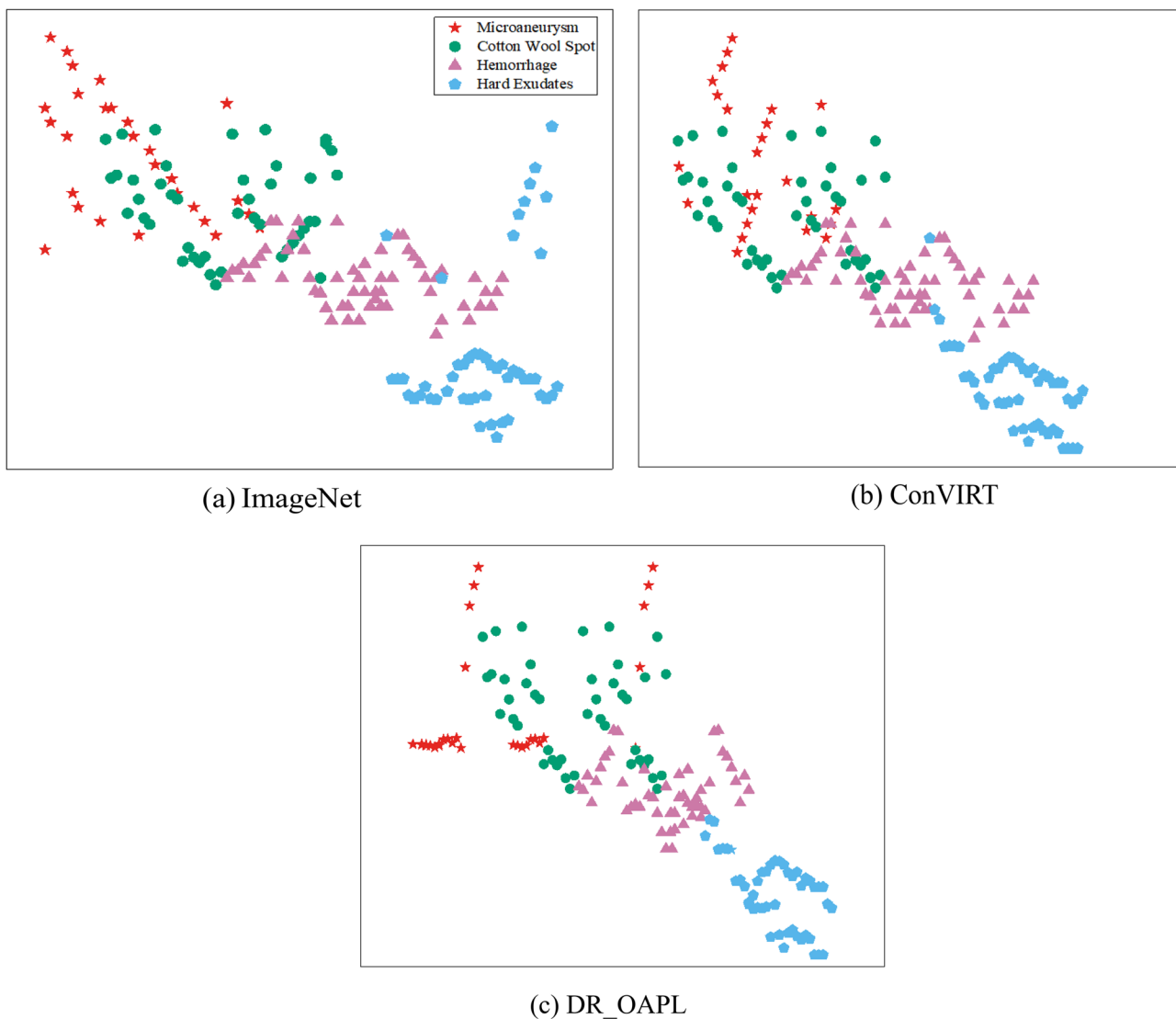


Fig. 5 t-SNE clustering diagram of different pre-training methods

Comparison with the transfer model learning ability To evaluate the effect of contrastive learning in DR_OAPL, we compared it with a transfer learning model, as shown in Fig. 6. The experiment was conducted on five internal test sets and one external test set. Positive values indicate that the contrastive learning model exhibits stronger learning ability compared to the transfer learning model. The results demonstrate that DR_OAPL achieves superior performance on the internal test sets, whether assessed at the image level or the patch level. This advantage is particularly pronounced in the external test set. This finding suggests that contrastive learning effectively uncovers common patterns and semantic information for both normal and abnormal lesions during the pre-training stage. By leveraging text embeddings and visual embeddings from auxiliary data, DR_OAPL optimizes performance on zero-shot detection tasks. The analysis demonstrates that DR_OAPL outperforms the transfer learning model across both image-level and patch-level evaluations on the internal test sets. This performance advantage is even more pronounced on the external test set. These results suggest that contrastive learning effectively identifies and utilizes common patterns and semantic features of both normal and abnormal lesions during the pre-training phase. By integrating text embeddings with visual embeddings from auxiliary data, DR_OAPL enhances its performance in zero-shot detection tasks,

highlighting the efficacy of contrastive learning in capturing relevant features for improved anomaly detection.

Evaluation of image encoders To evaluate the effectiveness of different image encoder pre-training methods, we compared three approaches: ImageNet (cross-domain), CLIP (cross-domain), and AnomalyCLIP (in-domain). The assessment utilized the MIMIC-CXR and CBIS-DDSM datasets, with lesion detection accuracy quantified by the area under the receiver operating characteristic curve (AUROC). The evaluation involved analyzing data effectiveness at various scales, with 1%, 10%, 50%, and 100% of the auxiliary data randomly extracted for each method. All reported results are the average AUROC scores from five independent runs, as detailed in Table 3.

Our findings indicate that CLIP, while also a cross-domain pre-training method, provides significantly superior image representation capabilities compared to ImageNet. Furthermore, the AnomalyCLIP pre-training method achieves the highest accuracy in image representation. Remarkably, AnomalyCLIP outperforms ImageNet even with just 1% of the training data, underscoring its effectiveness in overcoming challenges related to data privacy and scarcity in diabetic retinopathy (DR) detection tasks.

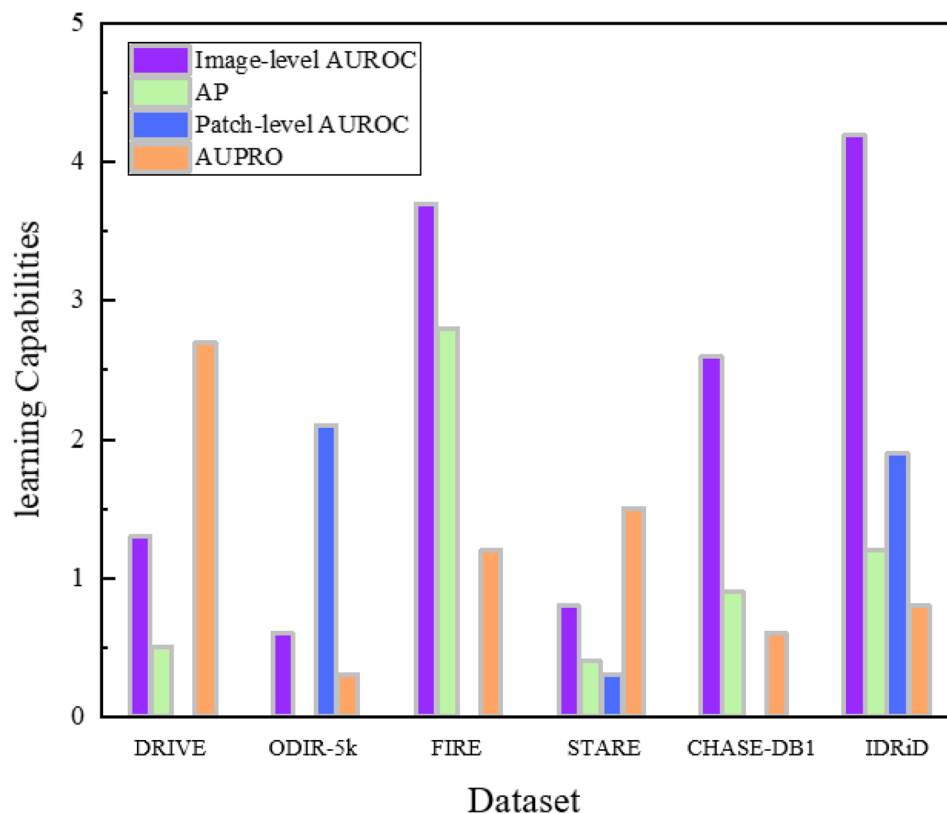


Fig. 6 The learning ability of DR_OAPL. (AP is the evaluation index at the image level, and AUPRO is the evaluation index at the patch level)

Table 3 Comparison of classification performance of different pre-training methods

Method	MIMIC-CXR				CBIS-DDSM			
	1%	10%	50%	100%	1%	10%	50%	100%
ImageNet	0.7971 $\pm$ 0.9	0.8283 $\pm$ 0.7	0.8541 $\pm$ 2.1	0.8695 $\pm$ 0.7	0.7357 $\pm$ 0.7	0.7930 $\pm$ 0.7	0.8229 $\pm$ 0.7	0.8531 $\pm$ 0.7
CLIP	0.8044 $\pm$ 0.6	0.8395 $\pm$ 1.5	0.8578 $\pm$ 1.2	0.8805 $\pm$ 0.1	0.7788 $\pm$ 0.8	0.8096 $\pm$ 0.4	0.8546 $\pm$ 1.3	0.8744 $\pm$ 1.6
AnomalyCLIP(Ours)	0.8669 $\pm$ 1.3	0.8758 $\pm$ 0.4	0.8935 $\pm$ 1.1	0.9057 $\pm$ 0.5	0.8555 $\pm$ 0.1	0.8721 $\pm$ 0.7	0.9001 $\pm$ 0.9	0.9241 $\pm$ 0.8

Table 4 Few-shot detection performance of each model

Method	MURA-1.1				CHASE-DB1			
	1	2	4	8	1	2	4	8
FCN	0.3323 $\pm$ 0.4	0.3583 $\pm$ 0.7	0.4541 $\pm$ 2.1	0.4995 $\pm$ 0.7	0.3375 $\pm$ 0.7	0.3930 $\pm$ 0.7	0.4227 $\pm$ 0.7	0.5531 $\pm$ 0.7
SegNet	0.4044 $\pm$ 0.6	0.4395 $\pm$ 1.5	0.4578 $\pm$ 1.2	0.5805 $\pm$ 0.1	0.3788 $\pm$ 0.8	0.4096 $\pm$ 0.4	0.4546 $\pm$ 1.3	0.5744 $\pm$ 1.6
U-Net	0.3669 $\pm$ 1.3	0.4758 $\pm$ 0.4	0.4935 $\pm$ 1.1	0.5057 $\pm$ 0.5	0.3555 $\pm$ 0.1	0.3721 $\pm$ 0.7	0.5001 $\pm$ 0.9	0.5241 $\pm$ 0.8
DeepLabV3+	0.3971 $\pm$ 0.9	0.4283 $\pm$ 0.7	0.5448 $\pm$ 2.1	0.5695 $\pm$ 0.7	0.3357 $\pm$ 0.7	0.3984 $\pm$ 0.7	0.4229 $\pm$ 0.7	0.5521 $\pm$ 0.7
DeepDR	0.4109 $\pm$ 0.6	0.4595 $\pm$ 1.5	0.4922 $\pm$ 1.2	0.5569 $\pm$ 0.1	0.3729 $\pm$ 0.8	0.4109 $\pm$ 0.4	0.4879 $\pm$ 1.3	0.5956 $\pm$ 1.6
DR_OAPL	0.7669 $\pm$ 1.3	0.7758 $\pm$ 0.4	0.7935 $\pm$ 1.1	0.8057 $\pm$ 0.5	0.7555 $\pm$ 0.1	0.7721 $\pm$ 0.7	0.8001 $\pm$ 0.9	0.8241 $\pm$ 0.8

Comparison with few-shot tasks Although the primary focus of DR_OAPL is its zero-shot transfer capability, its performance in few-shot learning scenarios is also significant, especially given the challenges of acquiring limited labeled data in the medical field. To evaluate this, we examined the impact of various few-shot settings on the model's detection performance during the fine-tuning stage and compared it to other transfer learning models. The specific few-shot settings included 1, 2, and 4 labeled samples, with results detailed in Table 4. The findings reveal that DR_OAPL retains strong lesion recognition ability even with limited labeled data. As the number of labeled samples increases, there is a corresponding improvement in model performance. Remarkably, DR_OAPL achieves better performance with just a single labeled sample than the best baseline model, which requires eight labeled samples. This demonstrates DR_OAPL's superior capability in task learning and its effectiveness in addressing the challenge of limited label acquisition in medical imaging.

Ablation study

Impact of different modules on performance To evaluate the impact of each module on diagnostic performance, we conducted ablation experiments, with results illustrated in Fig. 7. The base model (Base) serves as the reference for comparison. Module M1 incorporates the CONVIRT framework for comparison pre-training into the base model. Module M2 adds the Diagonally Prominent Attention Map (DPAM) to M1, while Module M3 introduces the Object-Agnostic Pre-Training (OAPT) based on M2. The results demonstrate a substantial improvement in DR_OAPL's performance with the integration of these modules, though the contribution of each varies. Adding Module M1 led to a 10.6% increase in the patch-level AUROC score and a 4% increase in the image-level AUROC score. This improvement underscores the effectiveness of contrastive learning in capturing the

alignment between local image features and text descriptions. The addition of Module M2 and subsequently Module M3 further enhanced performance. Specifically, incorporating the M3 module resulted in an increase of 12.2% in the patch-level AUROC score and 11.7% in the image-level AUROC score. This highlights the significant role of the OAPT module in improving diagnostic accuracy by enhancing lesion identification capabilities.

The impact of different context optimizations on performance The Object-Agnostic Pre-Training (OAPT) module is pivotal in our zero-shot detection experiments. This module enhances visual embeddings at both the patch and image levels by learning text embeddings, optimizing cosine similarity loss, and thereby capturing a range of lesion characteristics. This enhancement significantly improves the recognition capabilities of DR_OAPL. In designing the text instruction templates, we drew from previous research [19, 24] and tested various terms such as "Anomalous," "Damaged," and "Lesions." Our experiments revealed that the term "Lesions" yielded the most effective performance. The impact of global and local context optimization on DR_OAPL's performance was evaluated through an ablation study, as depicted in Fig. 8. In this study: "G" denotes global text optimization, "L" signifies local text optimization, "No-L&G" indicates the absence of both optimizers, "Only-L" represents local optimization alone, "Only-G" refers to global optimization alone, and "L&G" represents the integration of both optimizations. The results illustrate that incorporating both global and local context optimizations significantly enhances DR_OAPL's performance. Global optimization improves image-level semantic understanding, while local optimization refines patch-level details. The combination of these optimizations produces the most effective diagnostic results, highlighting the value of integrating both approaches in the model.

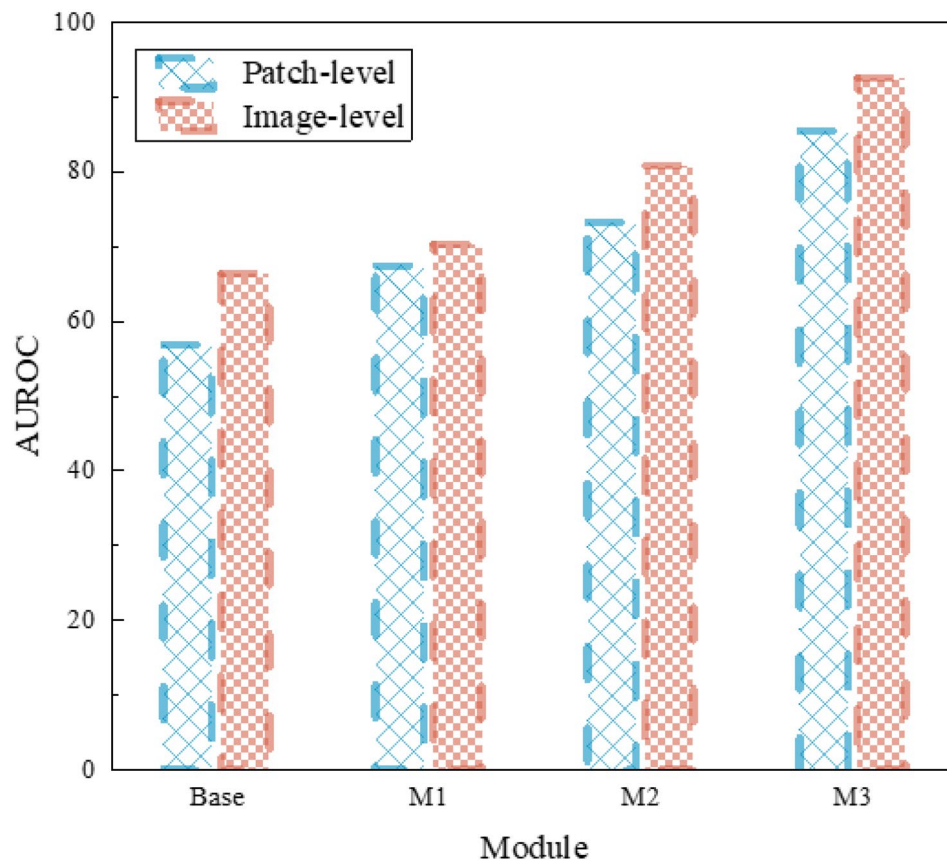


Fig. 7 AUROC scores of different modules

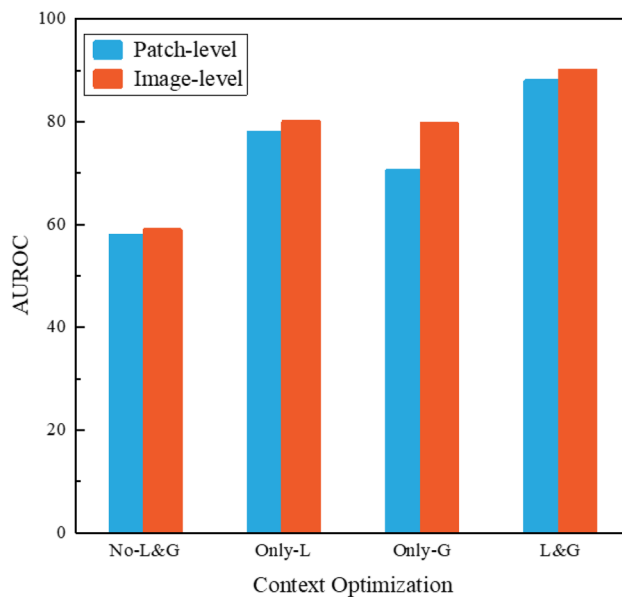


Fig. 8 AUROC scores for different context optimizations

Hyperparameter ablation We conducted an analysis of two key hyperparameters: the learnable text instruction length (L) and the learnable text instruction depth (D). The findings are illustrated in Fig. 9. For fine-tuning, we utilized the MURA-1.1 and CHASE-DB1 datasets to ensure robust parameter selection. Figure 9 presents the AUROC results at the image level in the first row and the AUPRO results at the patch level in the second row. The analysis revealed that the optimal values for both hyperparameters, across both datasets, are 14 for L and 9 for D. These values consistently yielded the best performance at both the patch and image levels. Consequently, we adopted L = 14 and D = 9 for all subsequent experiments.

Conclusion

To effectively locate and identify various types of diabetic retinopathy (DR), we developed an auxiliary diagnostic system, DR_OAPL. This system integrates a text instruction learning framework tailored for unknown targets through comparative pre-training and utilizes auxiliary data from the field to achieve zero-shot migration. DR_OAPL leverages both local (patch-embedding) and global (image-embedding) context information to enhance feature extraction within the visualization field. It employs

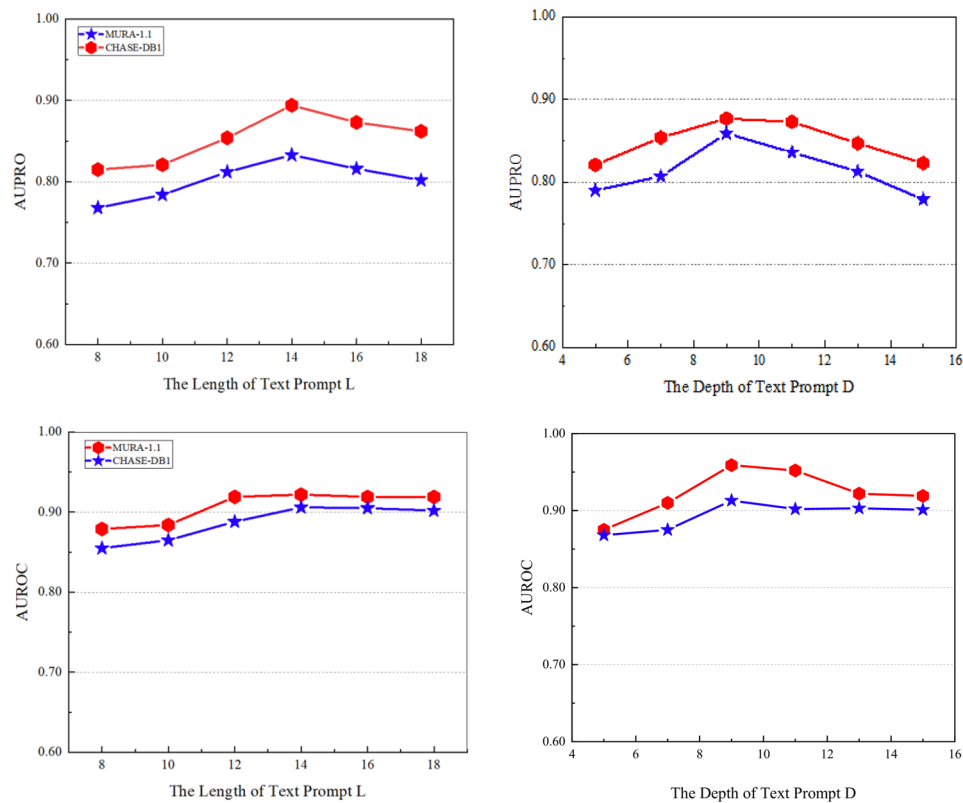


Fig. 9 Selection of hyperparameters. (The first row shows the AUROC results at the Image level, and the second row shows the AUPRO results at the Patch level)

contrastive learning to capture detailed feature representations of each lesion, which results in exceptional performance in DR detection. Compared to mainstream transfer learning methods, DR_OAPL offers significant improvements in detection performance and shows strong results in both zero-shot and few-shot tasks. This capability not only addresses privacy concerns related to medical data but also mitigates the challenges associated with costly and complex data annotation. Furthermore, DR_OAPL is particularly effective in detecting Microaneurysms, a critical early indicator of DR. Accurate detection of Microaneurysms is crucial for timely intervention and effective management of DR. Given the multimodality of medical data, applying DR_OAPL to multimodal data detection presents a promising avenue for future research. This study aims to provide valuable insights into related fields and contribute to advancements in medical imaging and diagnostics.

Abbreviations

DR	Diabetic retinopathy
t-SNE	T-distributed stochastic neighbor embedding
AI	Artificial intelligence
ZSL	Zero-shot learning
OCT	Optical coherence tomography
CFP	Color fundus photography
DL	Deep learning
NPDR	Non-proliferative diabetic retinopathy

VLMs	Visual language models
ZSAD	Zero-Shot Anomaly Detection
BERT	Bidirectional Encoder Representation from Transformers
CLIP	Contrastive Language-Image Pre-Training
ZSL	Zero-shot learning
VIT	Vision Transformer
ClinicalBERT	Clinical Bidirectional Encoder Representation from Transformers
ResNet50	Residual Network 50 layers
DPAM	Diagonally Prominent Attention Map
Q-K	Query-Key
Q-Q	Query-Query
K-K	Key-Key
V-V	Value-Value
CLAHE	Contrast-limited adaptive histogram equalization
AUROC	Area Under the Receiver Operating Characteristic Curve
AP	Average Precision
AUPRO	Area Under the Per-Region Overlap
DPAM	Diagonally Prominent Attention Map
DR_OAPL	Object-agnostic prompt learning for DR detection
CNNs	Convolutional neural networks
FCN	Fully Convolutional Networks

Supplementary Information

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Supplementary Material 1

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All authors contributed to the study conception and design. Xitong Guo and Xin Peng serve as corresponding authors of the article.

Author contributions

All authors contributed to the conception and design of the study. FeiFei Cao performed material preparation, data collection, and analysis. FeiFei Cao and ShuJu Li wrote the first draft of the manuscript. Xitong Guo, Meng Li, and Xin Peng conducted an experimental design on search stability, data imbalance issues, and revised the paper. All authors commented on previous versions of the manuscript. And all authors read and approved the final manuscript.

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Data availability

Public datasets links: Pre-training phase Datasets: (1) MIMIC-CXR: <https://www.physionet.org/content/mimic-cxr/2.0.0> (2) CBIS-DDSM: CBIS-DDSM - The Cancer Imaging Archive (TCIA) Public Access - Cancer Imaging Archive WikiFine-tuning phase Datasets: (3) MURA-1.1: <https://stanfordmlgroup.github.io/competitions/mura/> Test Datasets: (4) DRIVE: Introduction - DRIVE - Grand Challenge(5) ODIR-5k: <https://odir2019.grand-challenge.org/> (6) FIRE: FIRE: Fundus Image Registration Dataset (7) STARE: The STARE Project (8) CHASE-DB1: blogs.kingston.ac.uk/external-test-datasets/ (9) IDRiD: Home - IDRiD - Grand Challenge (10) EyePACS: <https://paperswithcode.com/dataset/kaggle-eyepacs> (11) RFMiD: <https://www.mdpi.com/2306-5729/8/2/29>.

Declarations

Ethics approval and consent to participate

No ethical approval was necessary for this study; all participants in this study consented to participation, and informed consent was obtained from all of the participants. The manuscripts reporting studies not involving human participants. All experiments in this study are based on public datasets and have no ethical implications. Our study does not adhere to the Declaration of Helsinki.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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